

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: BLURB | Context: EU/French Pharmaceutical Regulatory NLP — BLURB Assessment

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

BLURB's output ontology consists of task-specific label spaces: BIO/BIOUL token tags for NER over chemical/disease/gene/molecular-biology entity types [Q107, Q108, Q109], five-way ChemProt relation classes [Q65], yes/no and yes/maybe/no QA labels [Q86, Q90], binary cancer hallmark indicators [Q78, Q80], and a 0–4 BIOSSES regression score [Q74, Q75]. None of these label spaces accommodates regulatory-specific entity types (INNs, ATC codes, excipient nomenclature, posology expressions, EMA-template contraindication qualifiers) or regulatory-equivalence scoring. The BIOSSES regression operationalizes general biomedical semantic proximity ['estimated similarity score in the range from 0 (no relation) to 4 (equivalent meanings)' Q74] — not the legally determinative dose-threshold/population-qualifier sensitivity the deployment requires. Web-confirmed: no regulatory-equivalence STS benchmark exists [WEB-10], and 2025 SmPC LLM extraction shows ATC codes and excipient dosages are frequently misclassified [WEB-9]. The macro-averaged BLURB summary score [Q52, Q150] does not encode any decision-boundary or human-review-trigger semantics. Some weak structural overlap exists (binary classification, regression scoring) at the abstract level, hence score 2 rather than 1.

ID	Evidence
Q107	"BIO is the standard tagging scheme that classifies each token as the beginning of an entity (B), inside an entity (I), or outside (O)." (p.11)
Q108	"The NER tasks in BLURB are only concerned about one entity type (in JNLPBA, all the types are merged into one)." (p.11)
Q109	"Prior work has also considered more complex tagging schemes such as BIOUL, where U stands for the last word of an entity and L stands for a single-word entity." (p.11)
Q65	"We follow the ChemProt authors' suggestions and focus on classifying five high-level interactions — UPREGULATOR (CPR : 3), DOWNREGULATOR (CPR : 4), AGONIST (CPR : 5), ANTAGONIST (CPR : 6), SUBSTRATE (CPR : 9) — as well as everything else (false)." (p.8)
Q86	"The PubMedQA dataset [25] contains a set of research questions, each with a reference text from a PubMed abstract as well as an annotated label of whether the text contains the answer to the research question (yes/maybe/no)." (p.9)
Q90	"Each question is paired with a reference text containing multiple sentences from a PubMed abstract and a yes/no answer." (p.9)
Q78	"It contains annotation on PubMed abstracts with binary labels each of which signifies the discussion of a specific cancer hallmark." (p.9)
Q80	"We focus on predicting the top-level labels." (p.9)
Q74	"The Sentence Similarity Estimation System for the Biomedical Domain [54] contains 100 pairs of PubMed sentences each of which is annotated by five expert-level annotators with an estimated similarity score in the range from 0 (no relation) to 4 (equivalent meanings)." (p.9)
Q75	"It is a regression task, with the average score as the final annotation." (p.9)
Q52	"To compute a summary score for BLURB, the simplest way is to report the average score among all tasks. However, this may place undue emphasis on simpler tasks such as NER for which there are many existing datasets. Therefore, we group the datasets by their task types, compute the average score for each task type, and report the macro average among the task types." (p.7)
Q150	"The BLURB score is the macro average of average test results for each of the six tasks (NER, PICO, relation extraction, sentence similarity, document classification, question answering)." (p.14)
WEB-10	EMA 2024 AI Observatory Report — internal regulatory NLP tools developed under EMA staff oversight, not represented in any public benchmark (https://www.ema.europa.eu/en/documents/report/2024-ai-observatory-report_en.pdf)

WEB-9	2025 Frontiers SmPC IDMP study — ATC codes and excipient dosages 'frequently missing or misclassified' by LLMs (https://www.frontiersin.org/journals/medicine/articles/10.3389/fmed.2025.1598979/full)
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Strengths

- BLURB demonstrates that task-specific prediction layers can be cleanly layered atop a shared encoder [Q93, Q94, Q103, Q104], a transferable architectural pattern for regulatory tasks
- BIOSSES provides a concrete sentence-similarity regression schema with expert-derived gold scores [Q74, Q75] that could be structurally adapted into a regulatory-equivalence schema (with re-defined anchors)
- ChemProt's hierarchical relation taxonomy [Q65] is closer to a structured pharmaceutical interaction schema than other BLURB tasks

ID	Evidence
Q93	"Given an input token sequence, the language model produces a sequence of vectors in the contextual representation." (p.9)
Q94	"A task-specific prediction model is then layered on top to generate the final output for a task-specific application." (p.9)
Q103	"Many NLP applications can be formulated as a classification or regression task, wherein either individual tokens or sequences are the prediction target." (p.11)
Q104	"Modeling choices usually vary in two aspects: the instance representation and the prediction layer." (p.11)
Q74	"The Sentence Similarity Estimation System for the Biomedical Domain [54] contains 100 pairs of PubMed sentences each of which is annotated by five expert-level annotators with an estimated similarity score in the range from 0 (no relation) to 4 (equivalent meanings)." (p.9)
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Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output label categories are research-biomedical: chemical/disease/gene NER tags [Q108], generic chemical-protein relations [Q65], yes/no QA [Q86, Q90], cancer hallmarks [Q78]. None directly map onto regulatory NER categories (INNs, ATC, excipients, posology) or regulatory equivalence categories (dose-threshold mismatch, population-qualifier mismatch).

OO-2: Missing categories: INN entity type, ATC code entity type, excipient entity type, posology expression entity type, EMA-template contraindication qualifier; and on the STS side, regulatory equivalence with critical-mismatch sub-categories (dose, population, indication scope). The web search confirms no public benchmark covers these for French [WEB-10, WEB-12].

OO-3: BLURB's BIOSSES regression encodes 'general semantic proximity' [Q74] as the construct, which is value-laden in the sense that it weights paraphrase/synonymy uniformly — incompatible with the deployment's principle that small lexical differences (e.g., dose thresholds) carry legal weight disproportionate to their lexical distance.

OO-4: A stakeholder-driven taxonomy redesign is required: regulatory affairs experts under EMA/ANSM/EU AI Act normative framing [WEB-6, WEB-11] would need to define both the entity ontology and the equivalence-class taxonomy.

OO-5: Structural validity is harmed (the construct's structure — regulatory equivalence as legally-weighted proximity — is misrepresented as uniform semantic proximity); content validity is harmed (regulatory entity categories absent); external validity is at risk because BLURB scores cannot be expected to track regulatory compliance performance.

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Q90	"Each question is paired with a reference text containing multiple sentences from a PubMed abstract and a yes/no answer." (p.9)
WEB-10	EMA 2024 AI Observatory Report — internal regulatory NLP tools developed under EMA staff oversight, not represented in any public benchmark (https://www.ema.europa.eu/en/documents/report/2024-ai-observatory-report_en.pdf)
WEB-12	No regulatory-expert-annotated pharmaceutical NLP dataset (French or English) identified; DART Italian corpus does not use regulatory-expert annotators (https://arxiv.org/html/2510.18475)
Q74	"The Sentence Similarity Estimation System for the Biomedical Domain [54] contains 100 pairs of PubMed sentences each of which is annotated by five expert-level annotators with an estimated similarity score in the range from 0 (no relation) to 4 (equivalent meanings)." (p.9)
Q75	"It is a regression task, with the average score as the final annotation." (p.9)
WEB-6	EMA AI Reflection Paper requires human-oversight mechanisms and quality review for AI in product information drafting — implies threshold/calibration characterization the deployment must satisfy (https://www.ema.europa.eu/en/documents/scientific-guideline/reflection-paper-use-artificial-intelligence-ai-medicinal-product-lifecycle_en.pdf)
WEB-11	EMA AI page including LLM Guiding Principles (Sep 2024) (https://www.ema.europa.eu/en/about-us/how-we-work/data-regulation-big-data-other-sources/artificial-intelligence)

Information Gaps

- Whether the deployment's regulatory equivalence scoring should be ordinal, multi-label (per critical mismatch category), or hierarchical

Requires Expert Verification

- Definition of EMA/ANSM-aligned critical-mismatch categories and their decision thresholds for human-review triggering

Remediation

Gap 1: No label space for regulatory entity types (INN, ATC, excipient, posology, EMA-template qualifiers); BIOSSES regression encodes general semantic proximity, not regulatory equivalence [Q74, Q108].

Recommendation: Define a regulatory entity ontology aligned with WHO INN/ATC and EDQM Standard Terms, and define a regulatory-equivalence STS schema with explicit critical-mismatch sub-categories (dose threshold, contraindicated population, indication scope), informed by the deployment's documented critical mismatch list and the EMA AI Reflection Paper [WEB-6].

ID	Evidence
Q74	"The Sentence Similarity Estimation System for the Biomedical Domain [54] contains 100 pairs of PubMed sentences each of which is annotated by five expert-level annotators with an estimated similarity score in the range from 0 (no relation) to 4 (equivalent meanings)." (p.9)
Q108	"The NER tasks in BLURB are only concerned about one entity type (in JNLPBA, all the types are merged into one)." (p.11)

WEB-6

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